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Analysis of Dynamic Five-Element Model

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Abstract: As the theoretical basis of traditional Chinese medicine (TCM), five-element (Wu-Xing) model can generate abundant dynamic phenomena when it is considered as a dynamic system. In this paper, we extent the previous studied model and comprehensively consider all kinds of influence between elements, including "gain", "saturation", "generating", "is-generated", "restricting" and "is-restricted". Based on numerical and analytical cases study, we find that the system will converge to a fixed point or periodic solution, or become chaotic eventually under different settings. The stability is studied by using the first Lyapunov's method and the Lyapunov exponents, and it shows that all these cases are not stable. Additionally, by calculating the rate of change of the volume in phase space, we can judge this model is dissipative for "R&isR" model, but the model with "generation" involved can be dissipative under specific settings.

Key Words: five-element model, generation, restriction, periodic solution, fixed point, chaotic, dissipative

1 Introduction

Traditional Chinese medicine (TCM) is regarded as the typical wisdom of ancient Chinese. It has a holistic and systematic view of human and the whole world. The five-element (Wu-Xing) theory [1, 2], the theoretical basis of TCM, considers these five elements (metal, wood, water, fire and earth) are indispensable attributes in the world's construction. The relationships among them (see Fig.1) are mainly expressed as generation (also called "activation") and restriction (also called "inhibition"). Each of these relationships can be divided into two parts, and we take "fire" as an example: "$fire$" $\rightarrow$ "$earth$" ("fire" generates "earth", abbreviated as "G"), "$wood$" $\rightarrow$ "$fire$" ("fire" is generated by "wood", abbreviated as "isG"), "$fire$" $\dashrightarrow$ "$metal$" ("fire" restricts "metal", abbreviated as "R") and "$water$" $\dashrightarrow$ "$fire$" ("fire" is restricted by "water", abbreviated as "isR"). Nowadays there has been some diagnosis and treatment based on five-element theory, such as "judge the element is in either virtual or real state of Yin and Yang" and "fill his mother but rush down his son". If we can quantify the model from a mathematical perspective, it will be greatly significant to develop medicine research and system science.

The dynamic five-element model has abundant dynamic phenomena during the evolution, most of which can not be found in the static five-element model. Some mathematical approaches have been done to build and analyze the dynamic five-element system, which can be categoried as follows.

The five-element system can be built as discrete or continuous models. In discrete systems, states of elements can be quantified into *excess*, *normal* and *deficiency* [3], or expressed as *activity* and *inactivity* with boolen logic [4, 5]. Papers [6–8] use algebraic reasoning for relationships between elements. Based on the rules, the system will be deduced into one fixed point or one periodic solution [3–8]. While continuous systems establish corresponding differential equations and observe the process of the evolution of the system over time. As the continuous system might generate countless states in the time interval, it will eventually converge to one fixed point [9–13] or one periodic solution [13–15], or become chaotic [13, 14].

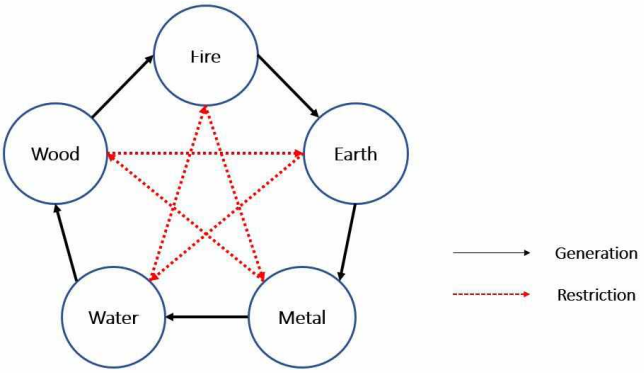


Fig. 1: The relationship among five elements.

Apart from four relationships mentioned, there exists another two terms: "gain" and "saturation", which mean the amount of the element can increase or decrease with the affect of the element itself respectively. Only "gain" and "isG" can increase the amount of the element while the others play an opposite role. In the disposal of interactions among five elements, "isG" and "isR" are the basis in all approaches. Except these two, papers [7–11, 13–16] take "gain" and "saturation" into account while [6–8, 11, 12, 17] consider relationships "G" and "R". There are only a few approaches [7, 8, 11] considering all six terms simultaneously in the dynamic five-element models. Particularly in continuous systems, the interactions between elements are set as linear terms (*e.g.* $\dot{x} = ax$) in [16, 17], but expressed as cross terms (*e.g.* $\dot{x} = bxy$) in [9–15].

In the study of system dynamics, there are three main

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aspects. One is to distinguish the final states of the system via the power spectrum, phase diagram or Lyapunov exponents [13, 14], or judging whether the system state changes after adding disturbance [5, 9, 10]. Another one is to analyze the stability of the system at equilibrium points, including the judgement whether real parts of eigenvalues of the linearized matrix are negative or not [12, 13], the trajectory diagram of variables [9, 10] and the use of the appropriate Lyapunov function [11]. The other is to obtain the general solution of a linear model [17], or by determining values of all parameters via biochemical data [15].

According to the above analysis, we build a continuous five-element system including all six terms. We analyze its dynamic characteristics by means of trajectory diagram, phase diagram and Lyapunov exponents. In our cases study, we find the systems can converge to fixed points or periodic solutions, or become chaotic, but they are always unstable, and their dissipativeness are related to the values of parameters, including the initial values of five elements as well.

2 Result

2.1 The five-element system

According to the theory of TCM, the value of the element could be influenced by any relevant relationship. For example, in the relationship "rebellion" (Wu), the restricted element might be counteractive at the one in the restricting size, which causes the reduction of both two elements. Therefore, we add the relationships "G" and "R" to the model proposed in the paper [9] and construct a more complete one as follows:

$$\begin{cases} \dot{x}_1 = a_1x_1 - b_1x_1^2 + c_1x_1x_5 - d_1x_1x_4 - g_1x_1x_2 - h_1x_1x_3 + e_1 \\ \dot{x}_2 = a_2x_2 - b_2x_2^2 + c_2x_2x_1 - d_2x_2x_5 - g_2x_2x_3 - h_2x_2x_4 + e_2 \\ \dot{x}_3 = a_3x_3 - b_3x_3^2 + c_3x_3x_2 - d_3x_3x_1 - g_3x_3x_4 - h_3x_3x_5 + e_3 \\ \dot{x}_4 = a_4x_4 - b_4x_4^2 + c_4x_4x_3 - d_4x_4x_2 - g_4x_4x_5 - h_4x_4x_1 + e_4 \\ \dot{x}_5 = a_5x_5 - b_5x_5^2 + c_5x_5x_4 - d_5x_5x_3 - g_5x_5x_1 - h_5x_5x_2 + e_5 \end{cases} \quad (1)$$

where x_i means the amount or state of ith element, $a_i, b_i, c_i, d_i, g_i, h_i$ are the coefficients of the relationship "gain", "saturation", "isG", "isR", "G" and "R" respectively and always nonnegetive, where e_i is the noise or intervention of ith element, $i = 1, 2, \cdots, 5$.

For simplicity, we consider $e_i = 0$ at the first step in our research and analyze how the system evolves in the stable environment. Additionally, we let $a_i = a, b_i = b, c_i = c, d_i = d, g_i = g, h_i = h$ for all i. Thus, equation set (1) can be changed into:

$$\begin{cases} \dot{x}_1 = ax_1 - bx_1^2 + cx_1x_5 - dx_1x_4 - gx_1x_2 - hx_1x_3 \\ \dot{x}_2 = ax_2 - bx_2^2 + cx_2x_1 - dx_2x_5 - gx_2x_3 - hx_2x_4 \\ \dot{x}_3 = ax_3 - bx_3^2 + cx_3x_2 - dx_3x_1 - gx_3x_4 - hx_3x_5 \\ \dot{x}_4 = ax_4 - bx_4^2 + cx_4x_3 - dx_4x_2 - gx_4x_5 - hx_4x_1 \\ \dot{x}_5 = ax_5 - bx_5^2 + cx_5x_4 - dx_5x_3 - gx_5x_1 - hx_5x_2 \end{cases} \quad (2)$$

This nonlinear system has strong coupling and symmetry. The plus and minus signs mean the positive and negative feedback respectively.

2.2 System dynamics analysis

In TCM, the values of five elements in any living body must be positive, so the values of the initial point are all positive, that is, $(x_1^0, x_2^0, x_3^0, x_4^0, x_5^0)$, $x_i(0) = x_i^0 > 0, i = 1, 2, \cdots, 5$.

The value of the variable cannot be less than zero. Otherwise, when the variable decreases from a positive value to a negative value, it will inevitably pass the value zero. According to equation set (2), $\dot{x} = 0$ holds, which means the variable will remain unchanged. It is impossible for the variable to continue to reduce to negative value. So the value of the variables must be non-negative during the evolution process.

The final state of the system can be generally classified into three types: fixed points, periodic solutions, and chaos. We have tested 300 randomly generated configurations of different parameters and initial values (all are positive), and we found that the system can converge to fixed points or periodic solutions, or become chaotic respectively.

In the following sections, we will give some examples respectively for fixed points, periodic solutions and chaotic cases with analysis of stability and dissipativeness.

2.2.1 Fixed points

In this section we give four cases which will converge to fixed points. The systems in the first two cases start evolution from the equilibrium points, while the other two evolve from non-equilibrium points.

Case 1. We set parameters to $a = b = c = d = g = h > 0$, and the solution to $\dot{x}_i = 0, i = 1, 2, \cdots, 5$, requiring all values are positive, is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

To judge the stability of the system at the equilibrium point, we use the first Lyapunov method, that is, linearize the original system at the equilibrium point, and then determine whether eigenvalues of the coefficient matrix contains positive real parts. The linearized matrix at the equilibrium point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is as follows:

$$Jaco = \begin{pmatrix} -1/3 & -1/3 & -1/3 & -1/3 & 1/3 \\ 1/3 & -1/3 & -1/3 & -1/3 & -1/3 \\ -1/3 & 1/3 & -1/3 & -1/3 & -1/3 \\ -1/3 & -1/3 & 1/3 & -1/3 & -1/3 \\ -1/3 & -1/3 & -1/3 & 1/3 & -1/3 \end{pmatrix}$$

The eigenvalues obtained from the above matrix are -1, 0.206-0.634i, 0.206+0.634i, -0.539-0.392i, -0.539+0.392i, which contain positive real parts, that means the system is unstable at the equilibrium point.

Case 2. We consider the system with only generating relationships "G" and "isG", denoted as "G&isG", which means $c > 0, g > 0, a = b = d = h = 0$. Then the equilibrium point only exists when $c = g$ and it is $(x_1, x_2, x_3, x_4, x_5) = (\delta, \delta, \delta, \delta, \delta), \delta > 0$. At this point the real parts of all eigenvalues of the coefficient matrix are zero. The system may be stable or unstable at this equilibrium point.

Case 3. We consider the system with only restricting relationships "R" and "isR", denoted as "R&isR",

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which means $d > 0, h > 0, a = b = c = g = 0$. Then the equation set (2) is reduced to:

$$\begin{cases} \dot{x}_1 = -dx_1x_4 - hx_1x_3 = -x_1(dx_4 + hx_3) \\ \dot{x}_2 = -dx_2x_5 - hx_2x_4 = -x_2(dx_5 + hx_4) \\ \dot{x}_3 = -dx_3x_1 - hx_3x_5 = -x_3(dx_1 + hx_5) \\ \dot{x}_4 = -dx_4x_2 - hx_4x_1 = -x_4(dx_2 + hx_1) \\ \dot{x}_5 = -dx_5x_3 - hx_5x_2 = -x_5(dx_3 + hx_2) \end{cases} \tag{3}$$

Starting from any configuration of $(x_1, x_2, \cdots, x_5)$ which are all positive values, the evolution of each variable is decreasing monotonically with time. Because the values of variables are always non-negative, each variable will eventually converge to a fixed point (such a case is shown in Fig. 2).

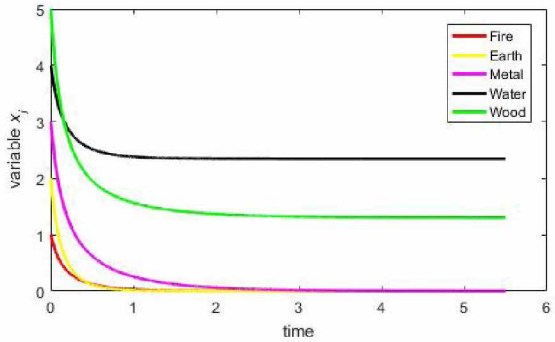


Fig. 2: System evolution over time with parameter settings of $(x_1^0, x_2^0, \cdots, x_5^0) = (1, 2, 3, 4, 5), d = h = 1, a = b = c = g = 0$.

Additionally, in this "R&isR" case there will be at least three variables that converge to zero. Otherwise, if convergent values of more than two variables are not zero, the restriction among them will make the variables continue to decrease until one of them converges to zero. At this point, the system is left with at most two variables that have no restricting connection, and whose values do not converge to zero.

To analyze the stability of the system, we introduce the Lyapunov exponent. The Lyapunov exponent judges the stability based on the degree of orbital divergence [18], and according to the paper [19], chaotic systems and systems with unstable periodic solutions have positive Lyapunov exponents, while the Lyapunov exponents of stable systems are all negative. We use singular value decomposition (SVD) method to solve the Lyapunov exponents [20]. We set time step to 0.001. The Lyapunov exponents are 0.0015, 0.0006, -1.3159, -2.3517 and -3.7092, two of them greater than zero. Thus, the system is convergent to a fixed point but unstable.

According to the paper [21], the rate of the change of the volume in phase space is defined as:

$$\frac{d}{dt}\int_\Omega \mathrm{d}\Omega = \int_\Omega \sum_i^N (\frac{\partial}{\partial q_i}\dot{q}_i + \frac{\partial}{\partial p_i}\dot{p}_i)\mathrm{d}\Omega \tag{4}$$

where q_i, p_i mean the generalized coordinate and the canonical momentum in the ith direction respectively, Ω means the volumn of phase space, which can be calculated as $\Omega = \int_\Omega \mathrm{d}\Omega = \int_\Omega \mathrm{d}q_1q_2\cdots q_Np_1p_2\cdots p_N$.

If $\frac{d}{dt}\int_\Omega \mathrm{d}\Omega > 0$, the volume of phase space will increase in the next instant, which means the trajectory may depart from the boundaries of the attractor; if $\frac{d}{dt}\int_\Omega \mathrm{d}\Omega < 0$, the volume of phase space will decrease, that is, the system is dissipative. In addition, if $\frac{d}{dt}\int_\Omega \mathrm{d}\Omega = 0$ holds all the time, it means the system is conservative and its trajectories lie on some surface of phase space. For case 3, the result is:

$$\begin{aligned} \frac{d}{dt}\int_\Omega \mathrm{d}\Omega &= \int_\Omega \sum_i^n \frac{\partial}{\partial x_i}\dot{x}_i \mathrm{d}\Omega \\ &= \int_\Omega -(d+h)(x_1+x_2+x_3+x_4+x_5)\mathrm{d}\Omega \\ &< 0 \end{aligned}$$

which shows the system is dissipative with an attractor.

Case 4. If the initial values of all variables are the same $x_1^0 = x_2^0 = x_3^0 = x_4^0 = x_5^0$, according to the strong symmetry of the equation set (2), the change trend of each variable is exactly the same. We only need to consider the following equation:

$$\begin{aligned} \dot{x} &= ax - bx^2 + cx^2 - dx^2 - gx^2 - hx^2 \\ &= x(a + (c - b - d - g - h)x) \end{aligned} \tag{5}$$

If $x(a + (c - b - d - g - h)x) \neq 0$, that is, x is not the equilibrium point, and then equation (5) would be changed into:

$$\begin{aligned} \frac{dx}{x(a + (c - b - d - g - h)x)} &= dt \\ (\frac{1}{x} - \frac{c - b - d - g - h}{a + (c - b - d - g - h)x})dx &= adt \\ ln|x| - ln|a + (c - b - d - g - h)x| &= at + k_1 \\ |\frac{a}{x} + (c - b - d - g - h)| &= e^{-at-k_1} \end{aligned} \tag{6}$$

Assuming the initial value is $x(t_0) = x_0, t_0 = 0$, we get the solution of the system(5) as follows:

$$x = \frac{ax_0}{Mx_0 + (a - Mx_0)e^{-at}}, \frac{a}{x} \neq M \tag{7}$$

where $M = b + d + g + h - c$.

When $t \to \infty$, we find that the system finally converges to a fixed value $\frac{a}{M} = \frac{a}{b+d+g+h-c}$, which is exactly the system's equilibrium point.

2.2.2 Periodic solutions

The case we consider in this section is: $(a, b, c, d, g, h) = (1, 1, 1, 1, 1, 1), (x_1^0, x_2^0, x_3^0, x_4^0, x_5^0) = (1, 2, 3, 4, 5)$, time step $\Delta_t = 0.001$. The simulation is as Fig. 3.

Based on the data generated by fourth Runge-Kutta (RK4), we can judge whether the variable has a period T in this integration interval according to the time interval between the maximum values of each variable. We find $T \approx 17.626$.

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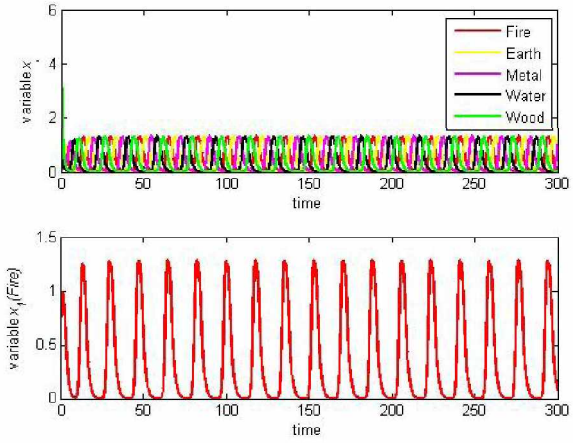


Fig. 3: System evolution over time with parameter settings of $(x_1^0, x_2^0, \cdots, x_5^0) = (1, 2, 3, 4, 5), a = b = c = d = g = h = 1$: the first picture describes the evolution of five elements, and the second picture describes the evolution of the element 'Fire'.

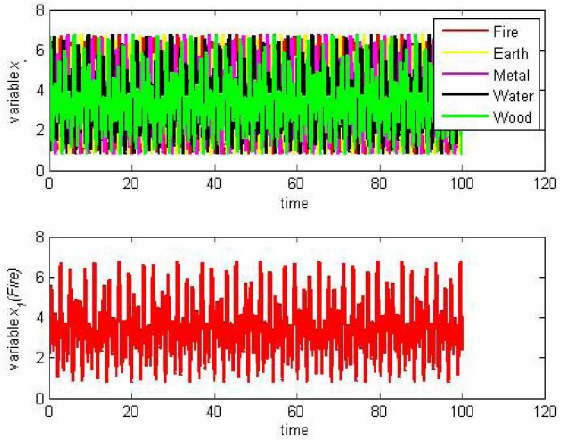


Fig. 4: System evolution over time with parameter settings of $(x_1^0, x_2^0, \cdots, x_5^0) = (1, 2, 3, 4, 5), c = g = 1, a = b = d = h = 0$: the first picture describes the evolution of five elements, and the second picture describes the evolution of the element 'Fire'.

Its Lyapunov exponents are 0.3885, -0.1719, -0.3473, -0.6833 and -0.9731. Thus, this system is convergent to an unstable periodic solution.

Additionally, the change of Ω is following:

$$\frac{d}{dt}\int_{\Omega} \mathrm{d}\Omega = 5\Omega - 4\int_{\Omega}(x_1 + x_2 + \cdots + x_5)\mathrm{d}\Omega$$

Whether the volume of phase space decreases or increases depends on whether $\frac{d}{dt}\int_{\Omega}\mathrm{d}\Omega < 0$ or not. However, based on the fact that the sum of all Lyapunov exponents is less than 0, the system is dissipative as a whole in the above case.

2.2.3 Chaos

The case we consider in this section is: $(a, b, c, d, g, h) = (0, 0, 1, 0, 1, 0), (x_1^0, x_2^0, x_3^0, x_4^0, x_5^0) = (1, 2, 3, 4, 5)$, time step $\Delta_t = 0.001$. The simulation of trajectory diagram is Fig. 4.

The period T cannot be found with RK4 and the trajectory in the phase diagram changes irregularly (Fig. 5), and the Lyapunov exponents of this system are 0.6700, 0.6606, -0.0761,-0.0794 and -1.1751, which indicate that the system is chaotic.

When $c = g > 0, a = b = d = h = 0$, the rate of change of Ω:

$$\frac{d}{dt}\int_{\Omega}\mathrm{d}\Omega = 0$$

which proves that the system is conservative with the fact that the sum of all Lyapunov exponents is zero, that is, no attractor exists in the system.

If we consider the case of $c > g > 0, a = b = d = h = 0$, which means the relationship "isG" is stronger than the relationship "G", the rate of change of Ω will be:

$$\frac{d}{dt}\int_{\Omega}\mathrm{d}\Omega = \int_{\Omega}((c-g)(x_1 + x_2 + x_3 + x_4 + x_5)\mathrm{d}\Omega \quad (8)$$

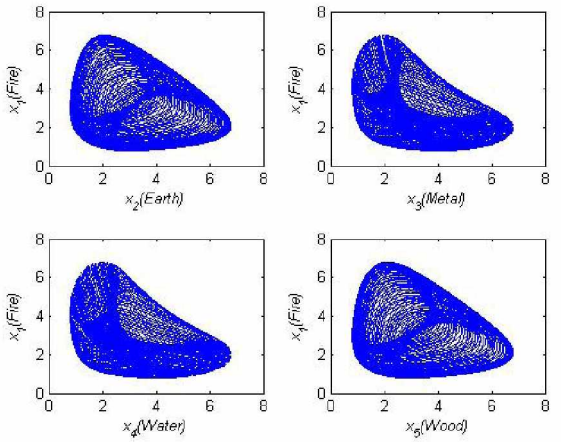


Fig. 5: Phase diagram of the relationship between element 'Fire' and other elements.

We have proved that the value of elements are all non-negative during the evolution, and all initial values are positive, so $\frac{d}{dt}\int_{\Omega}\mathrm{d}\Omega > 0$ in the equation (8) holds. The system will be absorbing energy and expanding its volume of phase space.

Similarly, the system is dissipative and forms an attractor if $g > c > 0, a = b = d = h = 0$, which means the relationship "G" is stronger than the relationship "isG".

2.3 Conclusion

The classic five-element model proposed by the ancients is a static description of the relationships among five elements. However, its applications to the human system and the natural laws of the world are dynamic. The ancients were limited by the level of mathematics at that time so that they could not describe the evolutionary dynamics of the five-element system, but today it is very important for us to establish a dynamic one. In this paper, we create a dynamic five-element model that in-

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cludes all six terms ("gain", "saturation", "generating", "is-generated", "restricting" and "is-restricted",) based on previous work [9], and according to the numerical simulation and mathematical analysis, we obtain the following conclusions:

- By case study, we find that the system may converge to a fixed point or a periodic solution, or become chaotic under different settings of parameters and initial values. In particular, we prove that the systems converge to a fixed point in the cases when there are only restricting relationships ($d > 0, h > 0, a = b = c = g = 0$) or initial values of five elements are equal ($x_1^0 = x_2^0 = \cdots = x_5^0$).
- Concerning the dissipativeness of system energy, we prove that: (1) when the five-element system just has restricting relationships (i.e., $d > 0, h > 0, a = b = c = g = 0$), it is dissipative and its energy is continuously decreasing; (2) when the five-element system just has generating relationships (i.e., $c > 0, g > 0, a = b = d = h = 0$), it is a conservative system if $c = g$, or a dissipative system if $c < g$, or its energy is increasing if $c > g$; (3) when the five-element system contains "gain" and "saturation" (i.e., $a > 0, b > 0$), the system may be dissipative, conservative or increasing energy depending on not only the initial values of five elements, but also values of all parameters.
- As for the stability, the cases studied in this paper are all unstable.

These results preliminarily reveal the complexity of the dynamic five-element model and its rich dynamic phenomena. Further analysis is needed to obtain a more systematic and complete theory. In addition, the construction of the model is still in theoretical speculation, and we should integrate it with the clinical application of TCM and keep imporoving it in the future.

References

[1] J. Tang,B. Liu,K. Ma, Traditional chinese medicine, The Lancet, 2008, 372(9654): 1938-1940.

[2] N. Mišić, Chinese concept of five phases (wu xing -yin yang) in the light of systems science, Proceedings of the First Symposium of Quantum-Informational Medicine QIM 2011: Acupuncture-Based & Consciousness-Based Holistic Approaches & Techniques, 2011: 118-127.

[3] J. Chen, Mathematical Modeling of Traditional Chinese Medicine, Journal of Mathematical Medicine, 2002 (06): 11-13.

[4] Q. Feng, X. Niu, S. Wang, Principles of Nonlinear Dynamics of Five Elements Theory, China Journal of Basic Medicine in Traditional Chinese Medicine, 2003 (7).

[5] P. Bellavite, M. Semizzi, S. Lussignoli, et al, A computer model of the 'five elements' theory of traditional Chinese medicine, Complementary Therapies in Medicine, 1998, 6(3):0-140.

[6] K. Meng, Mathematics of Yin-Yang and Five Elements and Its Application in Traditional Chinese Medicine, Acta Universitatis Traditionis Medicalis Sinensis Pharmacologiaeque Shanghai (6): 4-9.

[7] Y. Zhang, Mathematical Reasoning of Treatment Principle Based on "Yin Yang Wu Xing" Theory in Traditional Chinese Medicine, Chinese Medicine, 2011.

[8] J. Kim, M. Song, J. Kang, et al, A mathematical model for the deficiency-excess mechanism of yin-yang in five viscera, Chinese journal of integrative medicine, 2014, 20(2): 155-160.

[9] T. Wang, Research on the Dynamic Model of the WuXing Theory, Fudan University, 2008.

[10] G. Ding, T. WanG, Mathematical Analysis of Yin-Yang Wu-Xing Model in TCM, Journal of Acupuncture and Tuina Science, 2008, 6(5): 269-270.

[11] Q. Chen, A Mathematical Model of Yin-Yang Wu-xing Theory for Traditional Chinese Medicine, Proceedings of the 10th China Youth Information and Management Scholars Conference, 2008.

[12] Z. Yu, The Mathematical Model research about Traditional Chinese Medicine Science, Dalian University of Technology, 2005.

[13] X. Jiang, Study of Dynamics of a five-dimensional Nonlinear Network, Huazhong University of Science and Technology, 2006.

[14] F. Li, Functional structure model of human body and Yinyang-Wuxing equations, Physica Scripta, 1987, 36(6):966—969969.

[15] Q. Zhang, J. Han, Mathematical Simulation on Change Law of Wuzang Yinjing and Yangqi of Healthy Adults, Systems Engineering-Theory & Practice, 1998, 18 (7): 130-135.

[16] S. Chen, W. Zhang, One Yin-Yang Wu-Xing model of TCM, Journal of Acupuncture and Tuina Science, 2008, 6(5): 266-268.

[17] W. Zhao, Z. Zhao, The mathematical model of the theory of Yin Yang and five elements with the general solution of its derivation equation, Sichuan Journal of Traditional Chinese Medicine, 2006, 24 (11): 7-9.

[18] A. Wolf, J. Swift, H. Swinney, et al, Determining Lyapunov exponents from a time series, Physica D: Nonlinear Phenomena, 1985, 16(3): 285-317.

[19] C. Skokos, The Lyapunov characteristic exponents and their computation, Dynamics of Small Solar System Bodies and Exoplanets. Springer, Berlin, Heidelberg, 2010: 63-135.

[20] X. Zhang, Z. Li, L. Zhang, A Kind of Lyapunov Exponent Calculation Method Based on Singular Value Decomposition, Journal of University of Science and Technology Beijing, 2005, 27 (3): 371-374.

[21] C. Helrich, Analytical mechanics, Springer International Publishing, 2017, chapter 7.

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